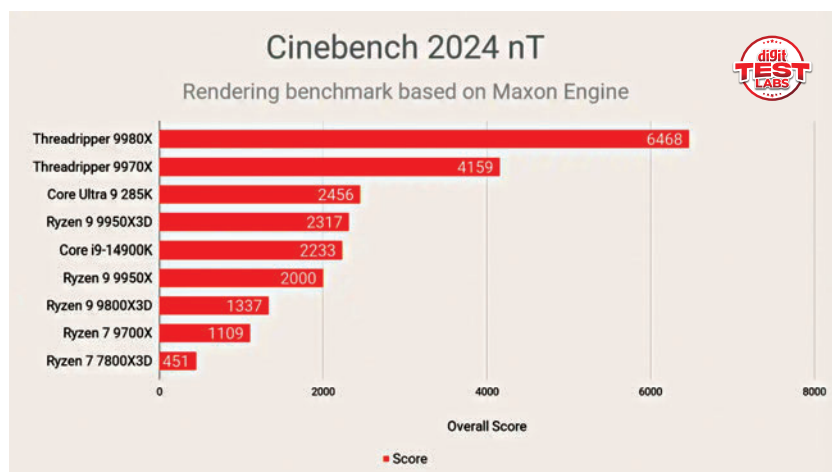




- **Memory:** 128 GB DDR5-6400 EXPO RDIMM (4 x 32 GB)
- **Storage:** The Crucial T705 1TB PCIe Gen 5 SSD
- **Power supply:** Corsair RM1200X Shift (1200W, ATX 3.0)
- **Operating System:** Windows 11 Pro, Version 24H2
- **Power Mode:** Best Performance

THERMALS

The Ryzen Threadripper 9980X is a dense, high-power processor and it behaves like one under sustained load. With a rated TDP of 350 watts, it requires serious cooling to maintain stable performance during prolonged rendering, compiling, or simulation sessions. In our open-bench setup, the SilverStone XE360-TR5 AIO cooler rose to the challenge, keeping the chip well within thermal limits throughout testing.

During an extended Cinebench R23 multi-core run, the CPU's average core temperature peaked at just 60.8°C. There were no signs of thermal throttling or instability. That's notably cooler than expected for a 64-core part, especially one running near its peak boost clocks. The result reflects both the efficiency of AMD's redesigned heatspreader and the effectiveness of the XE360-TR5's full-contact cold plate.

One area that stood out during stress testing was memory thermals. While the processor stayed cool, the RDIMMs, especially high-frequency modules, warmed up significantly under load.

With no active airflow directed at the memory, some modules approached thermal limits and risked throttling. Passive heatsinks alone weren't enough to mitigate this, even on an open bench. Anyone building a system around the 9980X should consider targeted airflow over the DIMM slots, particularly for memory-heavy workflows like simulation, 3D rendering, or AI inferencing.

Thermal performance was excellent in this configuration. That said, it's clear the 9980X benefits from thoughtful thermal planning. This isn't a CPU that belongs in a budget chassis with average airflow; it needs a well-ventilated case, a capable cooler, and memory-specific thermal consideration to perform at its best. Do that, and it rewards you with stable clocks, high throughput, and consistent all-core performance.

POWER CONSUMPTION

If performance defines half of the Threadripper 9980X's character, power draw completes the picture. In light workloads such as browsing, document editing, and routine multitasking, the system stayed relatively modest, drawing about 100 watts in total, which is expected for a high-end desktop equipped with a modern GPU.

When pushed, though, the 9980X shows its appetite. The CPU package power peaked at 349.7 watts, almost exactly in line with its TDP, confirming that AMD's specification is not just theoretical. Even when idle, the system

maintained a steady baseline near 85 watts which is an acceptable result given the number of active cores and background services.

What stands out, though, is not just the raw draw, but how well the power translates into meaningful work. The 9980X delivers substantially higher throughput than older HEDT parts like the Core i9-10980XE or even the previous-gen 3990X, and it does so with greater efficiency per watt. The system remained stable throughout thanks to the 1200 Watt PSU. If you're running a workstation of this calibre, a high-efficiency PSU is essential.

AMD RYZEN THREADRIPPER 9980X VERDICT

The AMD Ryzen Threadripper 9980X is a heavyweight. Built for creators, engineers, and high-performance professionals, it combines 64x Zen 5 cores, high boost clocks, and an I/O-rich TRX50 platform to deliver staggering levels of compute power.

What sets it apart from previous Threadrippers and even some enterprise chips is its flexibility. The 9980X handles heavily threaded workloads with ease, but thanks to its 5.4 GHz boost clock and refined IPC, it doesn't flinch when you throw single-threaded or lightly threaded tasks at it. This makes it surprisingly competent in interactive workflows like editing, design, and software development, where low latency matters as much as throughput.

—Mithun Mohandas

SPECIFICATIONS

Architecture: Zen 4 | Cores/Threads: 64/128 | Base Clock: 3.2GHz | Boost Clock: 5.4GHz | L2/L3 Cache: 64MB/256MB | PCIe 5.0 Lanes: 80 | Memory Support: DDR5-6400 | ECC Support | Overclocking Unlocked | TDP: 350 W | Socket: sTR5 | Boost Algorithm: Precision Boost 2 | Max Temperature (TJMax): 95°C

CONTACT

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